

公開仕様から見るLogカーブ - 静的図版

微分・2階微分・接続条件 / メーカー公開文書の式を再実装した比較

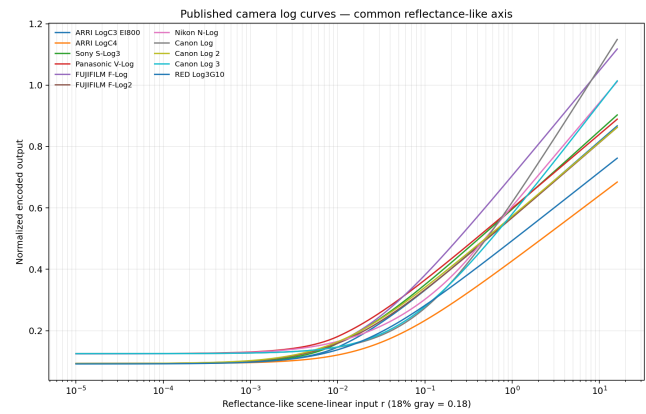
このPDFは、インタラクティブHTMLとは別系統の静的図版です。

全体図は現在の比較図をそのまま残し、その後に各カーブの個別コンタクトシートを収録しています。

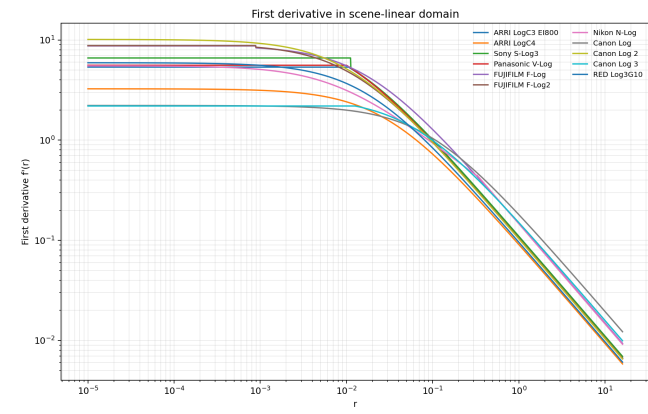
留保：ここで扱う各社Logは、仕様書・データシート・ホワイトペーパーに記載された公開数式の再実装です。実機測定によるOETF推定ではなく、内部画像処理、センサー固有処理、EI依存処理、量子化、クリップ等を再現するものではありません。公表係数の丸めにより、接続点の微小な差が設計上の不連続か丸め誤差か区別できない場合があります。

個別シート：上段 = scene-linear全域の f , f' , f'' 。下段 = 接続点近傍の拡大。破線は公開式の接続点です。

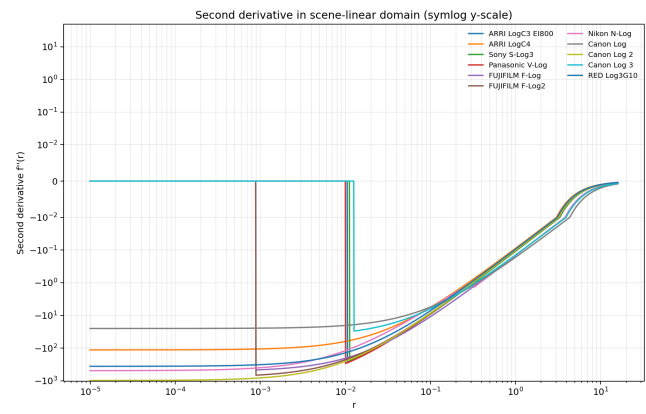
Overview - 全カーブ比較



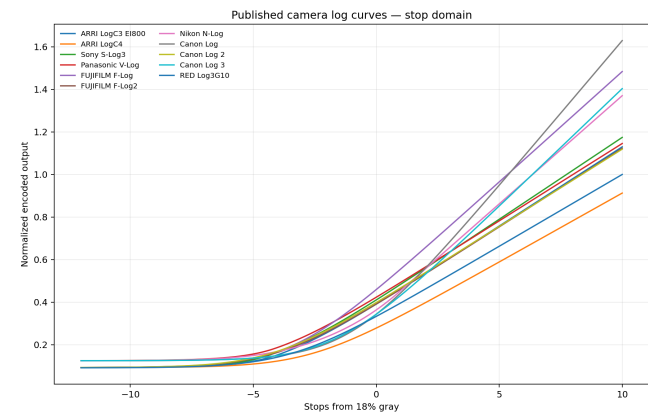
scene-linear: curve



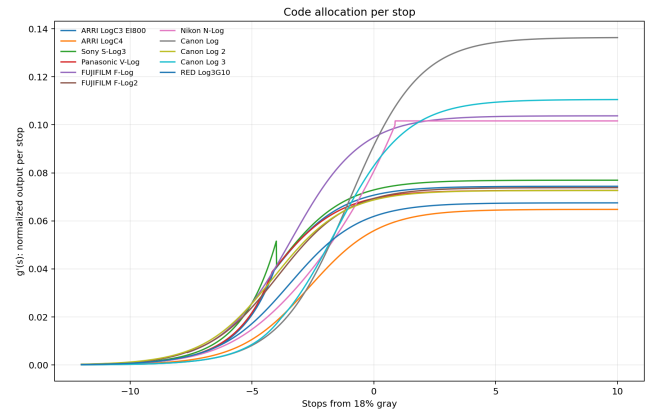
scene-linear: first derivative



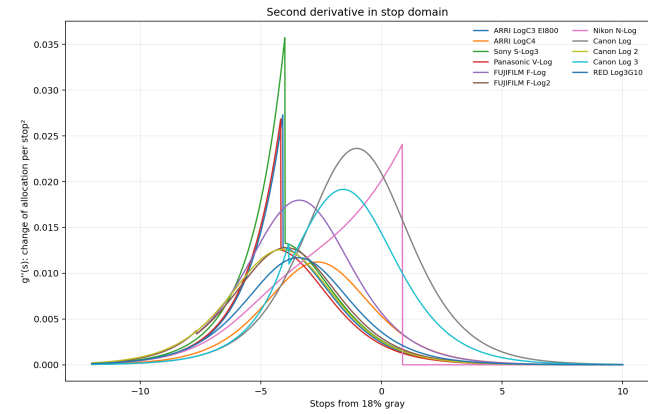
scene-linear: second derivative



stop domain: curve



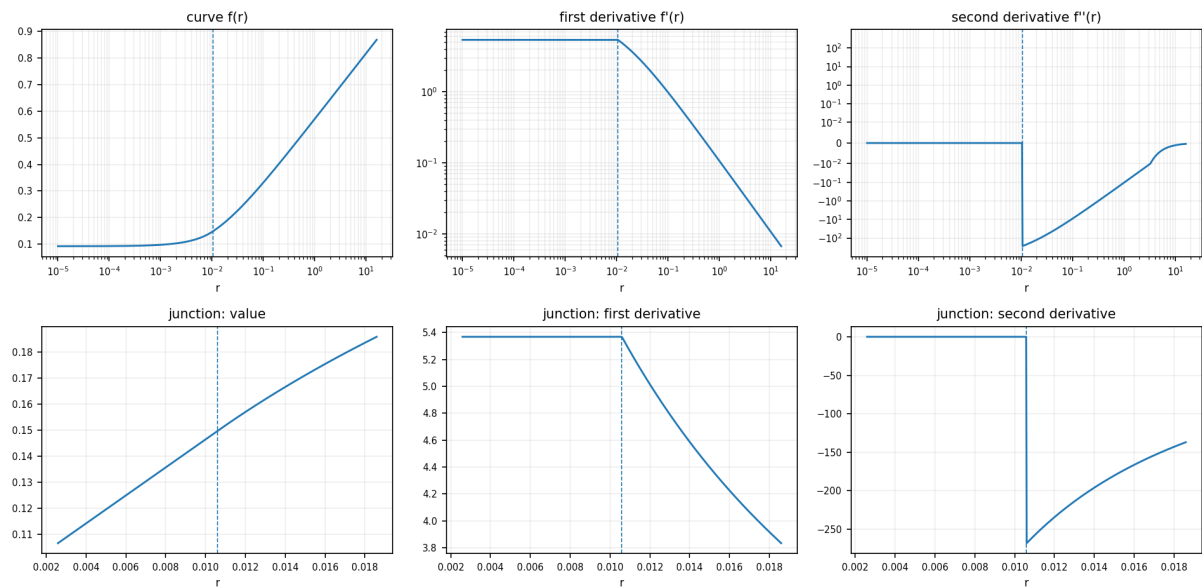
stop domain: first derivative



stop domain: second derivative

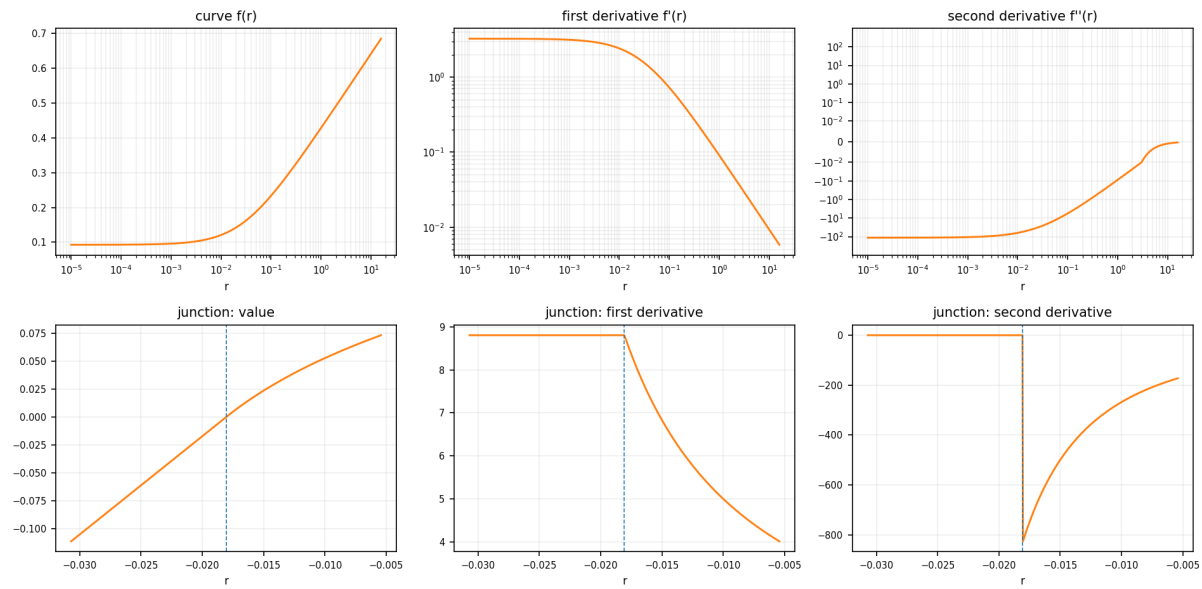
Individual curve contact sheets

ARRI LogC3 EI800



Published-formula implementation; source key: arri-logc3

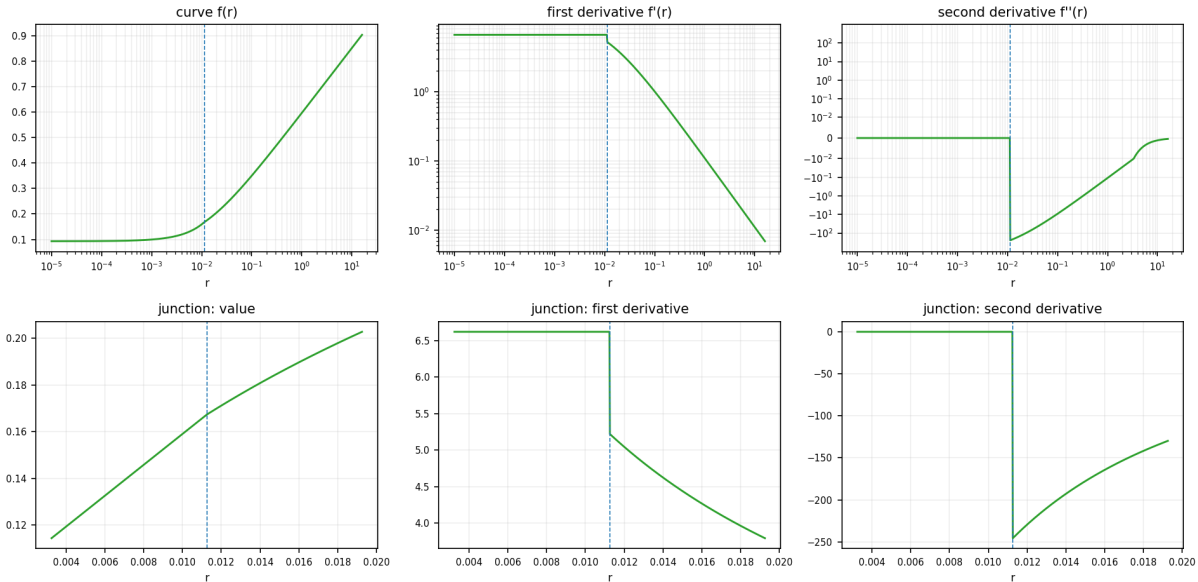
ARRI LogC4



Published-formula implementation; source key: arri-logc4

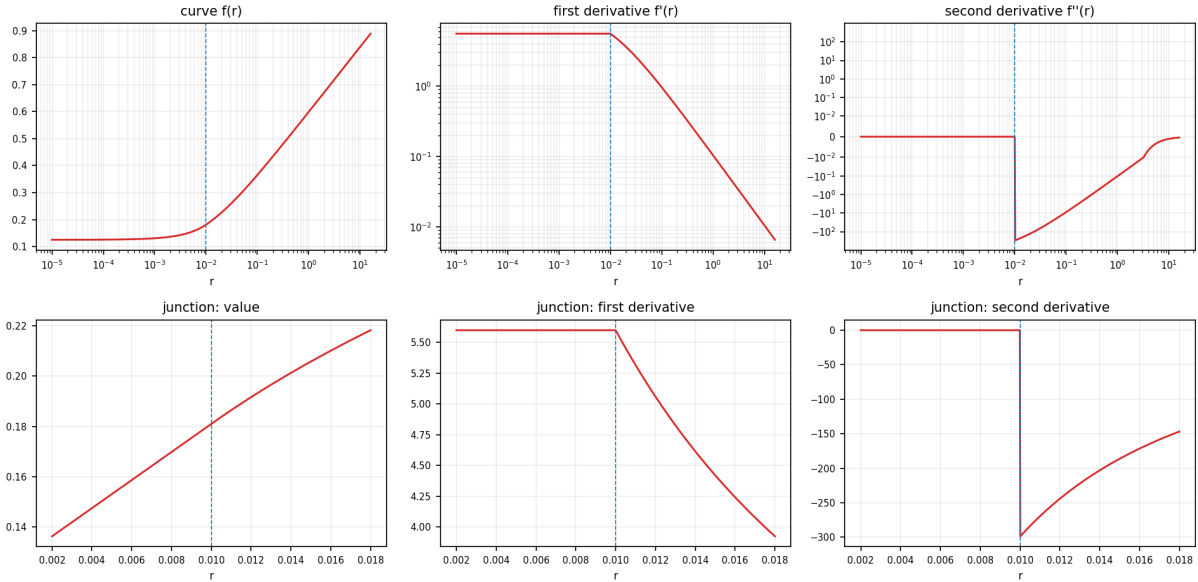
Individual curve contact sheets

Sony S-Log3



Published-formula implementation; source key: sony-slog3

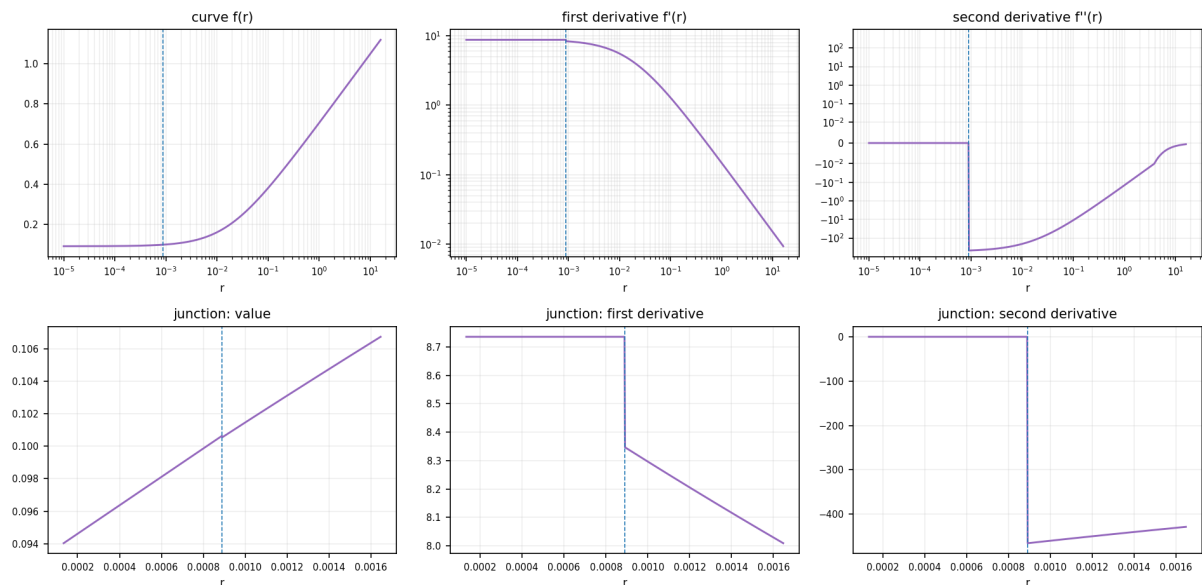
Panasonic V-Log



Published-formula implementation; source key: panasonic-vlog

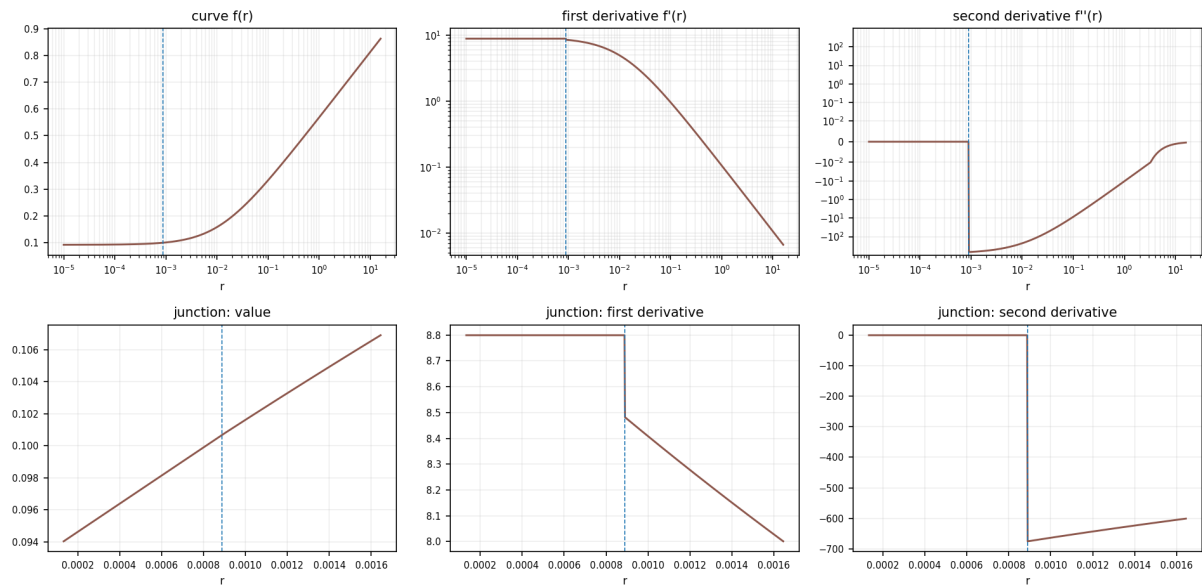
Individual curve contact sheets

FUJIFILM F-Log



Published-formula implementation; source key: fujifilm-flog

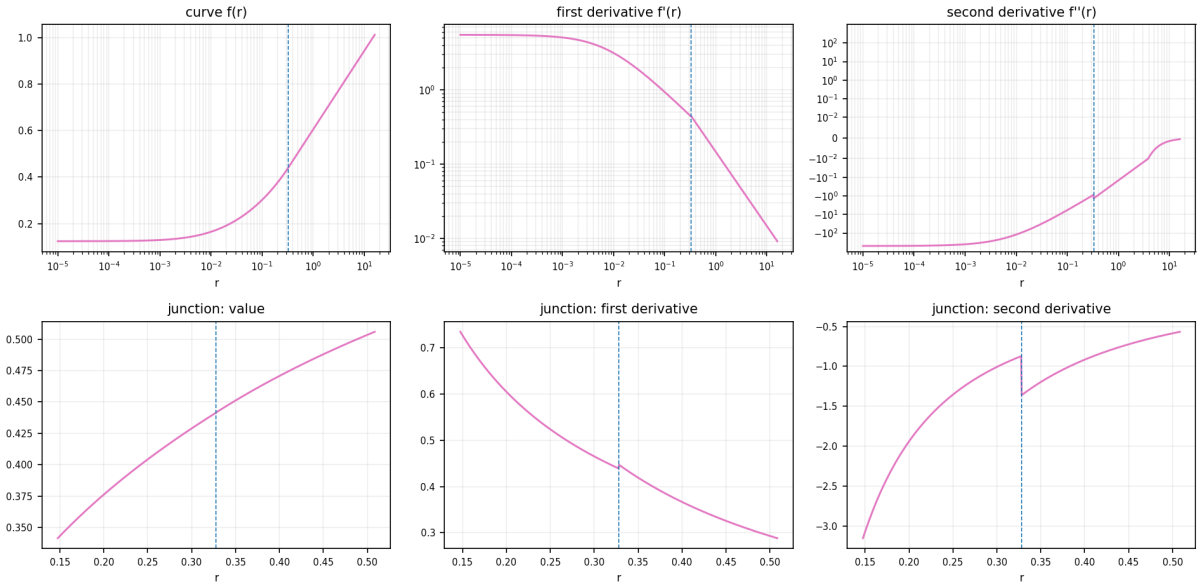
FUJIFILM F-Log2



Published-formula implementation; source key: fujifilm-flog2

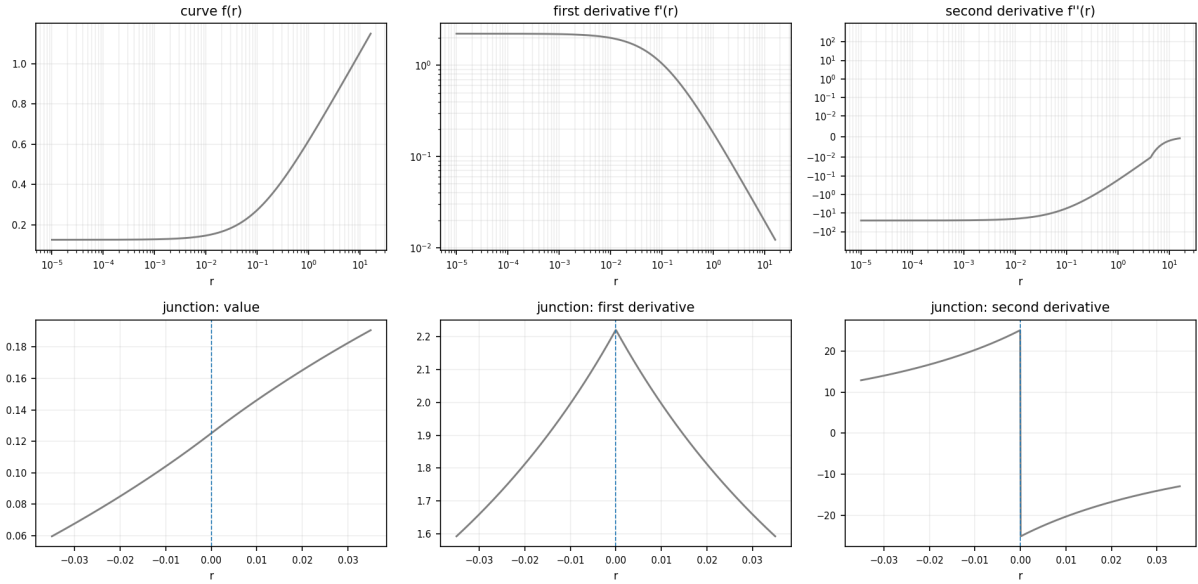
Individual curve contact sheets

Nikon N-Log



Published-formula implementation; source key: nikon-nlog

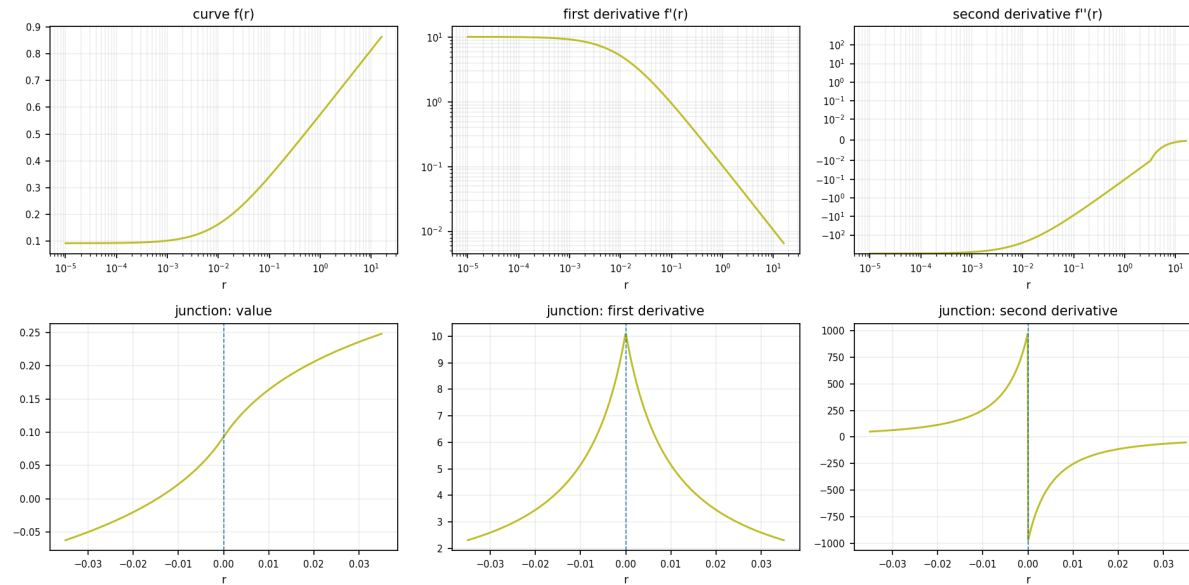
Canon Log



Published-formula implementation; source key: canon-log

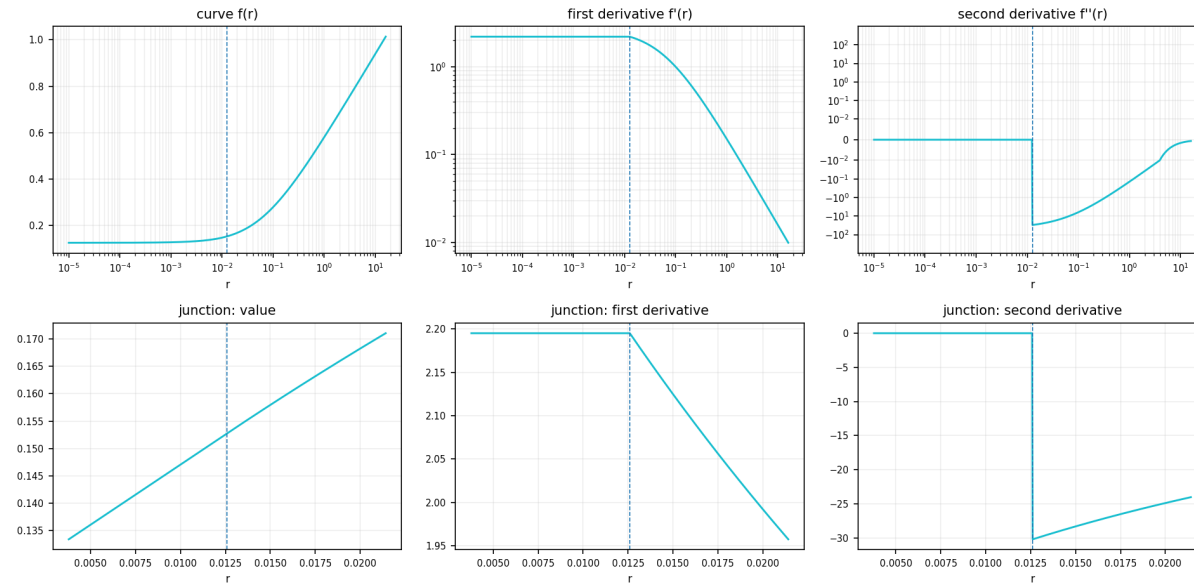
Individual curve contact sheets

Canon Log 2



Published-formula implementation; source key: canon-log

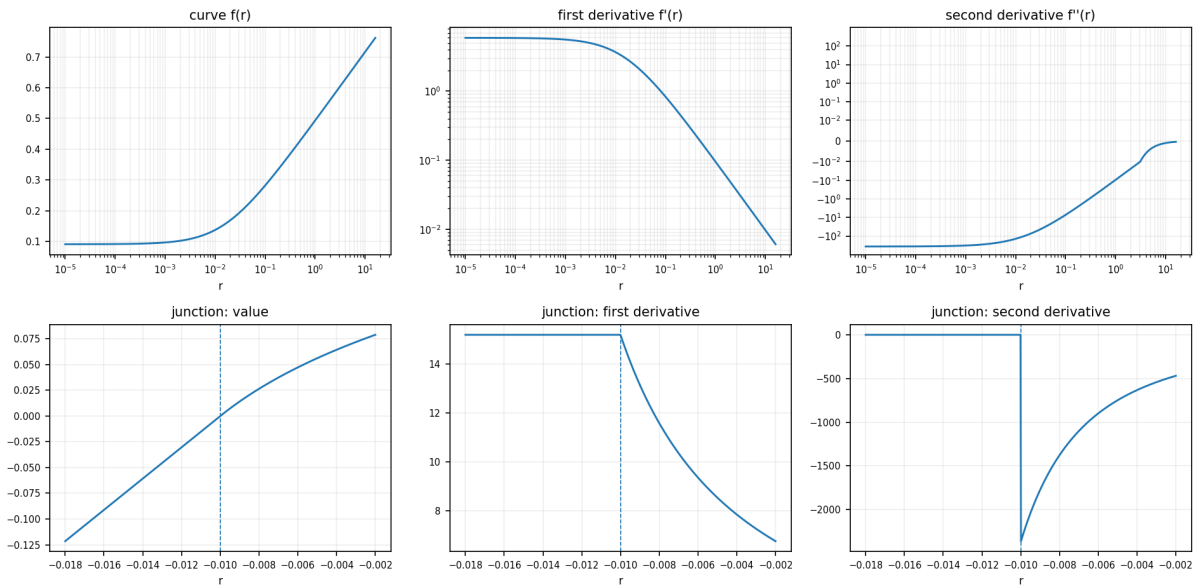
Canon Log 3



Published-formula implementation; source key: canon-log

Individual curve contact sheets

RED Log3G10



Published-formula implementation; source key: red-log3g10

参考文献 / implementation sources

各カーブの数式を拾った公開文書。CIE L*とCineonは比較参照。

ARRI LogC3	ARRI, ALEXA Log C Curve: Usage in VFX, 2017.
ARRI LogC4	ARRI, ARRI LogC4 Logarithmic Color Space Specification, 2025.
Sony S-Log3	Sony Corporation, Technical Summary for S-Gamut3.Cine/S-Log3 and S-Gamut3/S-Log3.
Panasonic V-Log	Panasonic Corporation, V-Log/V-Gamut Reference Manual Rev.1.0, 2014.
FUJIFILM F-Log	FUJIFILM Corporation, F-Log Data Sheet Ver.1.2, 2024.
FUJIFILM F-Log2	FUJIFILM Corporation, F-Log2 Data Sheet Ver.1.1, 2024.
Nikon N-Log	Nikon Corporation, N-Log Specification Document Ver.1.0.0, 2018.
Canon Log family	Canon USA, Canon Log Gamma Curves white paper, 2018.
RED Log3G10	RED Digital Cinema, White Paper on REDWideGamutRGB and Log3G10, Rev C.
CIE L*	CIE, CIE 1976 L*a*b* colour space; see also ISO/CIE 11664-4.
Cineon	Eastman Kodak, Cineon Digital Film System technical documentation; reference implementation cross-checked with FilmLight technical note.

URLと完全な書誌情報は companion QMD / references.bib に収録しています。